CHM33900
Biochemistry: A Molecular Approach
Spring 2026

Instructor **Professor Angeline Lyon**
email: lyonam@purdue.edu

Lectures are in BHEE 129 on Mondays, Wednesdays, and Fridays from 9:30-10:20 am. Recorded lectures and slides will be available on Brightspace.

Brightspace is your home for everything related to CHM339. Check it often to stay current!

TAs	Ketaki Mahurkar	kmahurka@purdue.edu
	Alli Bajmakovich	abajmako@purdue.edu
	Christian Guess	guess0@purdue.edu
	Sofia Schumann	schumans@purdue.edu
	Ethan DeHaan	edehaan@purdue.edu
	Maximillian Ante	mante@purdue.edu
	Matthew Schmidt	schmi340@purdue.edu
	Mikayla Bell	bell401@purdue.edu
	Lea Vojslavek	lvojslav@purdue.edu

Office Hours Office hours are held weekly. If you are unable to attend the scheduled office hours, email Dr. Lyon or the TAs, and include "CHM339" in the subject line. Emails will be replied to within 24 hours.

Dr. Lyon	Wednesday	11:30 am – 12:30 pm	WTHR 104
Ketaki Mahurkar	Monday	10:30 am – 11:30 am	BRWN B106
	Friday	10:30 am – 11:30 am	BRWN B106
Alli Bajmakovich	Tuesday	1:30 pm – 2:30 pm	BRWN B106
Christian Guess	Tuesday	3:00 pm – 4:00 pm	BRWN B106
Sofia Schumann	Thursday	10:30 am – 11:30 am	BRWN B106
Ethan DeHaan	Tuesday	10:30 am – 11:30 am	BRWN B106
Maximillian Ante	Thursday	12:15 pm – 1:15 pm	BRWN B106
Matthew Schmidt	Monday	11:30 am – 12:30 pm	BRWN B106
Mikayla Bell	Monday	12:30 pm – 1:30 pm	BRWN B106
Lea Vojslavek	Monday	2:30 – 3:30 pm	BRWN B106

Course Description This comprehensive one-semester biochemistry course emphasizes molecules and molecular reaction mechanisms, building upon the principles conveyed in general chemistry and

organic chemistry. It is designed to cover the essential elements of biochemistry traditionally covered in a two semester series. Topics to be covered include: amino acids, peptides and proteins, molecular models and structure-function relationships, enzymes, enzyme kinetics and enzyme mechanisms, carbohydrates and carbohydrate metabolism, aerobic metabolism, lipids and membranes, lipid metabolism, nitrogen metabolism and integration and regulation of metabolism. Special emphasis will be placed on connections between biochemical principles and the fields of medicine, human health and disease, nutrition and biotechnology. This course should serve as excellent preparation for higher-level biochemistry courses as well as standardized pre-health professional exams. CHM33900 was designed as the 4th semester of the 1-2-1 competency-based curriculum transformation in chemistry as it applies to pre-health professional and life science students at Purdue. This course is also open to students who meet the prerequisites.

Learning Resources, Technology, and Texts

Recommended textbook: *Biochemistry Concepts and Connections*
Appling, Anthony-Cahill, and Mathews
1st Edition (2016) or 2nd Edition (2018)

Lectures will be recorded in BoilerCast and posted on Brightspace within 24 hours of class. All course information, including recorded lectures, homework assignments, quizzes, exams, and keys will be made available through Brightspace.

Recommended app: Aktiv Chemistry

This year, we will be using the Aktiv app for general and organic chemistry review, in class questions, and practice problems.

Learning Outcomes The goals of this class are for you to:

1. Learn the structures and properties of biological molecules.
2. Understand the basics of how cells do the chemistry that is essential for life.
3. Understand the fundamental metabolic pathways essential for life and their regulation.
4. Interact productively with your peers on these topics throughout the semester.

Student Responsibilities We want you to be successful in this class, and will do our best to help you. However, we expect you to hold up your end of the bargain by attending and participating in every class and help/review session, speaking up if there are questions or issues, accessing the resources available to you (office hours, etc), and completing all assignments and any extra credit opportunities as provided. Check the course web site often as it will be updated frequently.

If you find yourself struggling with the material, please come and get help from Prof. Lyon and the TAs *early*. We are here to help you successfully navigate the course, and can help you best if you stay on top of the assignments and material throughout the semester. This course covers a lot of information, and builds on topics covered at the beginning of the semester. Ask questions early and often to ensure you have a strong foundation.

Assignments:

Five	Homework Assignments (30 points each)	150 points
Three	Semester Exams (125 points each)	375 points
One	Cumulative Final Exam (2 x 125 points)	250 points
Pre-drop Total		775 points

You are allowed to drop **ONE** exam (one semester exam or half the final) – 125 points

TOTAL **650 points**

Homework Assignments There will be **five** problem sets during the semester that must be turned in via Gradescope by 11:59 pm on the following days: **January 26, February 9, March 6, March 30, and April 24**. Assignments will be made available on Brightspace at least one week in advance of the due date. You may and are encouraged to discuss these problem sets freely with your peers. However, you must compose and write your own independent answers. Papers with identical or nearly identical answers, or answers plaigerized from other sources, will be given an automatic zero. Late assignments will automatically lose 30% of the points, and will not be accepted once the answer key has been posted, one week after the due date.

AI Policy. AI programs, such as ChatGPT, can be a useful tool. In the context of this course, AI tools may be used for assistance on homework assignments, practice problems, and additional information on course topics. *There is no guarantee that the outputs will be correct*. I strongly discourage their use in the course, but if you choose to use them, do so at your own risk.

If you choose to use AI for homework assignments, *you must include in your submission the program you used and inputs you used with your questions*.

The exams will test your comprehension and understanding of the material, as well as how to apply it. Relying on AI is highly unlikely to help you be successful on exams.

Exams There are three exams scheduled during the semester:

February 11, 2026 (Wednesday) at 8 pm in BHEE 129 and PHYS 112

March 12, 2026 (Thursday) at 8 pm in CL50 224 and FRNY G140

April 8, 2026 (Wednesday) 8 pm in WTHR 200 and WALC 1055

All exams are **in-person** from 8-9:30 pm on the dates and locations listed above. Room assignments will be announced in classes beginning the week before the exam and posted on Brightspace.

Any exams that have answers that are identical, nearly identical, or that have been plaigerized will be given an automatic zero.

To account for the time taken by exams, there will be **no lectures on February 13, March 13, and April 10.**

Missed Exams As a general rule, we do not give makeup exams. If you have a documented illness or or University-approved absence, we will work with you on accomodations. Otherwise, if you miss an exam, it will automatically count as your dropped exam grade.

Re-Grade Policy You have **one week** from the day the key is posted to submit a regrade or recount request for homework and exams. Homework regrade requests must be submitted through Gradescope. Exam regrades must be submitted in writing, along with the exam, to Dr. Lyon or one of the TAs. Keep in mind that your entire assignment will be regraded, and your score may go up, stay the same, or go down. There are **no exceptions** to this policy.

If you find an addition error, the assignment is NOT subject to regrading. Points will only be recounted.

Final Exam University policy on Final Exams states: *Students scheduled for more than two (final) examinations in one calendar day are entitled to reschedule any examination in excess of two. It is the*

responsibility of the student to make necessary arrangements ***before the last week*** *of regularly scheduled classes.*

If you are ill or have an emergency during the final exam, you will first need written documentation of the illness (e.g., a note from a physician stating that you were incapacitated due to illness or documentation of the emergency) and have that documentation verified by the Dean of Students Office. Once receiving proper documentation, send that information to Dr. Lyon.

Your exam will be prorated as follows: you will automatically drop half of the final instead of any midterm exam we already had. The three exam scores from the semester will be averaged, and will count as half of the final.

Incompletes A grade of incomplete (I) will be given only in unusual circumstances. To receive an 'I' grade, a written request must be submitted prior to May 1 and approved by the instructor. The request must describe the circumstances, along with a proposed timeline for completing the course work. Submitting a request does not ensure that an incomplete grade will be granted. If granted, you will be required to fill out and sign an "Incomplete Contract" form that will be turned in with the course grades. Any requests made after the course is completed will not be considered for an incomplete grade.

Grading Scale: Final grades will be based on the following cutoffs:

A+	97-100%
A	90-96.9%
B+	87-89.9%
B	80-86.9%
C+	77-79.9%
C	70-76.9%
D+	65-69.9%
D	55-64.9%
F	0-54.9%

Academic Integrity It is expected that all students will uphold the highest standards of academic integrity. The expectations are as follows:

1. All graded material must be your own independent effort. Plagiarism in any form is unacceptable. Do not allow your work to be copied by other students.
2. You may discuss the take-home problem sets freely with other students. However, you must compose and write your own independent answers.
3. Your answers on all assignments must be in your own words.
4. Turning in an altered assignment, quiz, or exam for re-grading is a violation of academic integrity.
5. The minimum penalty for academic dishonesty of any sort is a score of zero on the assignment, quiz, or exam in question.

The Office of the Dean of Students (ODOS) states that the "appropriate standard of proof is based upon a preponderance of evidence." For example, any quizzes or exams with identical or nearly identical answers, or answers that have been plaigerized from other sources, will be submitted to ODOS for independent investigation, and possible further action. The penalty may be a grade of "F" for the course, or further action. Additional information is available through Brightspace University Policies and Statements "Purdue's Student Guide for Academic Integrity" link.

Attendance Policy This course follows the Academic Regulations: Attendance and Office of the Dean of Students: Class Absences posted in Brightspace under "University Policies and Statements." The policies state that students are expected to be present for every meeting of the classes in which they are enrolled. When conflicts or absences can be anticipated, such as for many University-sponsored activities and religious observations, you should inform me of the situation as far in advance as possible. For unanticipated or emergency absences when advance notification to is not possible, contact me as soon as possible by email. For cases that fall under excused absence regulations, you or the your representative should contact or go to the Office of the Dean of Students (ODOS) website to complete appropriate forms for instructor notification. Under academic regulations, excused absences may be granted by ODOS for cases of grief/bereavement, military service, jury duty, parenting leave, or emergent medical care.

Course Evaluation During the last two weeks of the semester, you have the opportunity to give feedback on this course and your instructor using an online course evaluation system. You will receive an official email from evaluation administrators with a link to the online evaluation site and will have up to 10 days to complete this evaluation. Your participation is an integral part of the success of this course. I read all of the evaluations, and am continuously revising the course to take your suggestions into account.

Nondiscrimination Statement We are committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her own potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life.

Accessibility Purdue University strives to make learning experiences as accessible as possible. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let me know so that we can discuss options. You are also encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247. More details are available on our course Brightspace under Accessibility Information.

Mental Health:

- **If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try WellTrack.** Sign in and find information and tools at your fingertips, available to you at any time.

- **If you need support and information about options and resources**, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are M-F, 8 am- 5 pm.

- **If you find yourself struggling to find a healthy balance between academics, social life, stress**, etc. sign up for free one-on-one virtual or in-person sessions with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is completely free and can be done on BoilerConnect. If you have any questions, please contact Purdue Wellness at evans240@purdue.edu.

- **If you're struggling and need mental health services**: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help,

such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS office of the second floor of the Purdue University Student Health Center (PUSH) during business hours.

Emergency Preparation In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructors or TAs via email. You are expected to read your @purdue.edu email on a frequent basis.

Basic Needs Security Any student who faces challenges securing their food or housing and believes this may affect their performance in the course is urged to contact the Dean of Students for support. There is no appointment needed and Student Support Services is available to serve students 8 a.m.-5 p.m. Monday through Friday. Students may submit requests for emergency assistance from the Critical Needs Fund.

Disclaimer This syllabus is subject to change. You will be notified of any changes as far in advance as possible via an announcement on Brightspace. Monitor your Purdue email daily for updates.

Course Schedule

Topic	Readings (1st Ed.)	Semester Week
General Chemistry and Organic Chemistry Review Material	Ch. 1 – 3 Online General and Organic Chemistry Primer	Weeks 1 – 2
Amino Acids, Peptides & Proteins	Ch. 5 – all sections Ch. 6 – all sections	Weeks 3 – 6
Enzymes & Enzyme Kinetics	Ch. 8 – sections 8.1– 8.9	Weeks 6 – 7
Carbohydrates	Ch. 9 – all sections	Weeks 7 – 8
Lipids, Membranes, and Cellular Transport	Ch. 10 – all sections	Weeks 8 – 9
Introduction to Metabolism	Ch. 11 – all sections	Weeks 9 – 10
Carbohydrate Metabolism: Glycolysis, Gluconeogenesis, Glycogen Metabolism & the Pentose Phosphate Pathway	Ch. 12 – all sections	Weeks 10 – 11
The Citric Acid Cycle	Ch 13 – all sections	Weeks 11 – 12
Electron Transport and Oxidative Phosphorylation	Ch 14 – sections 14.1 – 14.7	Weeks 12 – 13
Lipid Metabolism	Ch. 16 – sections 16.1 – 16.2	Weeks 13 – 14
Amino Acid and Nitrogen Metabolism	Ch. 18 – all sections	Weeks 14 – 15

Chapter 17 will also be covered and incorporated throughout Lecture Topics 6 – 11: *"Interorgan and Intracellular Coordination of Energy Metabolism in Vertebrates"*